The Successful Facilitation of Virtual Team Meetings

Brooke B. Allison

Marissa L. Shuffler

&

Allison M. Wallace

Department of Psychology,
Clemson University

Allison, B. B., Shuffler, M. L., & Wallace, A. M. (2015). The successful facilitation of virtual team meetings. In J. A. Allen, N. Lehmann-Willenbrock, & S. G. Rogelberg (Eds.), The Cambridge handbook of meeting science (pp. 680– 706). New York, NY: Cambridge University Press.

Address correspondence to:
Marissa L. Shuffler, Ph.D.
Department of Psychology,
Clemson University
418 Brackett Hall
Clemson, SC 29634
(864) 656-2831
mshuffl@clemson.edu

Abstract

Virtual teams, whose members may be engaged in interdependent tasks while geographically dispersed, are highly prevalent workgroups within organizations today. Because these global virtual teams must rely on technology to fulfill team goals across time and space, such teams may warrant unique meeting styles and structure. The following chapter reviews the literature on virtual team meetings, providing evidence regarding the factors which may aid in the facilitation of such meetings based on existing research, and offers practitioners and professionals guidelines for successfully facilitating virtual team meetings. Drawing upon previous meetings and global virtual team research, those wishing to conduct a virtual team meeting are encouraged to consent to the following recommendations: select a facilitator, select appropriate information and communication technology, set meeting norms, set and reinforce team roles, acknowledge time zone and cultural differences, and follow up with action items.

Key words: virtual meetings, global virtual teams, virtual teamwork, meeting facilitation, meeting inputs

The Successful Facilitation of Virtual Team Meetings

"My supervisor has just stopped by to remind everyone to meet upstairs in 10 minutes. Today is Friday—team meeting day. I finish preparing my weekly summaries, lock the company desktop, and head to the conference room with my notes. When I arrive, I join my team at the table and face the black teleconferencing system in the center of it. Our team is geographically dispersed across several states within the US, with the slight majority of teammates being collocated at my location. I collaborate mainly with a woman from New York who will be dialing into today's meeting from home. My supervisor adjusts the volume on the sound system and I now recognize the voice of another coworker speaking to us from the black box; she is met with enthusiasm and thanked by all present for her work well-done this week. We wait for the others to dial in while my supervisor reminds those of us in the room to call up the electronic agenda he emailed us yesterday. Side chatter and small talk happens among those of us in the room. When the others join, we greet them, my supervisor goes over the agenda, and the meeting begins with each member updating the team on his or her current role and how their work is impacting the team's goals. Time flies quickly in our meetings. Soon, the hour has nearly passed and our supervisor is encouraging the group to ask questions about the tasks we have just gone over for the following week. About an hour after the meeting, one of our teammates emails the group a meeting summary and a series of action items to complete based on what we've just discussed."

Introduction

The brief recollection described above by one of our authors demonstrates a somewhat ideal example of how team meetings can be conducted successfully in today's virtually connected workplace. This is particularly true when the appropriate structures and interventions are offered to those involved in such meetings, such as the group trainings offered to those

participants within this example. Virtual teaming plays a significant role in today's organizational landscape, especially in the form of virtual teams (VTs), where team members are geographically dispersed across different locations, culturally diverse, and working together utilizing a range of technologies to communicate and share information (Gibson & Cohen, 2003). Due in part to the diverse expertise and 24/7 market strategy that is allowed by virtual teaming, cooperate conglomerates and small businesses alike have adopted and placed a high value on virtual teams. Luckily, many employees have also adapted to interacting with coworkers and supervisors electronically over the years, as have many organizations, such as those who now have separate rooms solely devoted to virtual meetings (Beise, Evairsto, & Niederman, 2003).

The meeting previously described in our example occurred via teleconference, but many other common types of communication modes exist for virtual team meetings, such as videoconferencing, group decision making software, and web conferencing, with varying levels of quality for both content and social interactions (Kirkman & Mathieu, 2005). Other common aspects of virtual team meetings not captured by this case study concern those of differing cultures and technologically-related differences (Von Glinow, Shapiro, & Brett, 2004). How team members react to and handle these virtual team meeting challenges may affect both the proximal outcomes of any given meeting as well as the long term impact at both team and organizational levels (Kauffeld & Lehmann-Willenbrock, 2012).

Those involved in successful global virtual teams (GVT) may find that they prepare for meetings often and in advance, demonstrating how pertinent preplanning is to virtual team process and outcomes (Niederman & Volkema, 1999). It has been remarked that "the role of the team facilitator is to move the team towards its objectives" (Pauleen & Yoong, 2001, p. 192),

and, though this applies to all types of teams, the role of a successful team facilitator of a virtual team may call for increased social engineering and acknowledgement. Individuals placed on virtual teams are typically thought to be invaluable to the organization in terms of expertise (Malhotra, Majchrzak, & Rosen, 2007) and, therefore, make important contributions to group meetings. However, these talented individuals also discharge unique cognitions and behaviors, making the onboarding and maintenance processes of VT meetings that much more important.

Although research and practice have converged over the past several decades in order to better align advances in technology with organizational outcomes, notably in the product design and health arenas (Karsh, 2004: Sanchez & Mahoney, 1996), more could certainly be done in organizational settings to promote successful VT meetings. Further, there is still much to explore in terms of research regarding such meeting preparation and facilitation. The current chapter therefore offers a review of the relevant meetings and virtual teamwork literatures with a focus on translating scientific findings into practical guidelines for the successful facilitation of such meetings. This chapter is intended to provide useful information to both potential facilitators and members of virtual teams, and important factors to consider include information and communication technology (ICT), meeting norms, team member roles, and time zone and cultural differences as they related to virtual team meetings. Further, this chapter is designed to offer future research directions regarding where research may be critically needed in relation to preparing for VT meetings. We begin by focusing on one particular type of meeting, the initial kickoff meeting held by VTs, as such meetings are particularly critical and must be especially attended to in terms of facilitation. Next, we offer guidelines for meeting practices based on the extant literature in this area. Finally, we conclude with several future research directions that may lend to a better understanding of how to best conduct such meetings.

The Importance of the Initial Meeting in VTs

Whereas every team meeting is important, the first team meeting plays a crucial role in setting the tone for all remaining meetings and interactions (Coutu, 1998; Suchan & Hayzak, 2001). Many of the issues that may arise in virtual teams are likely to occur if team members do not set the right tone with one another during their initial interactions; indeed, if initial meetings are not conducted effectively, they may lead to lower levels of trust (Alge, Wiethoff, & Klein, 2003), poorly developed shared mental models (Burke, Shuffler, Salas, & Gelfand, 2010), and lower overall effectiveness (Warkentin & Beranek, 1999). We thus first highlight this specific component of virtual team meetings given that the subsequent guidelines may be particularly critical to consider for the initial GVT meeting.

Many virtual project teams begin with some type of kickoff or pre-brief meeting that is held in a face-to-face setting (Hertel, Geister, & Konradt, 2005). When these meetings follow effective face-to-face meeting approaches and norms, such as appropriate communication behaviors, procedural characteristics, and temporal characteristics, they may aid in initiating the team development cycle and enhance collaborative processes, such as communication and coordination of roles and responsibilities (Cascio & Shurygailo, 2003; Leach, et al., 2009). Because there is evidence to suggest online communication and other types of virtual communication create specific challenges for users, including lack of clarity into others' emotions or motives (Culnan & Markus, 1987; Rice & Love, 1987), holding initial meetings face-to-face may provide a forum by which clarity can be gained and an identification with the team can be created in order to better facilitate later meeting interactions (Wisenfeld, Raghuram, & Garud, 1999).

Although some companies may view initial in-person meetings as counterproductive and/or unnecessary expenses for virtually-operating teams, science and practice continually demonstrate that face-to-face meetings at the onset of team formation are essential to building personal relationships and trust among team members (Henttonen & Blomqvist, 2005; Holton, 2001), facilitating participative decision making and mitigating conflict (Powell, Piccoli, & Ives, 2004), increasing team satisfaction (Ocker, Fjermestad, Hiltz, & Johnson, 1998), and allowing for an understanding of individual communication styles to enable future virtual meetings (Pauleen & Yoong, 2001). Importantly, initial face-to-face meetings allow team members to put a face with the person they will be communicating with in the future, aiding in the development of trust and social relationships that may otherwise be challenging in virtual environments (Olson & Olson, 2002). Having a mental image of a face to associate with a voice on the telephone or the text typed on an instant messaging chat screen may alleviate the psycho-technological barrier for some individuals that would otherwise perceive lack of human contact to be a work stressor.

Although effective initial and interval face-to-face meetings cannot fully compensate for a lack of informal interaction and meetings experienced among collocated teams once a project begins, they do offer important benefits, such as helping to establish and clarify roles, goals, and expectations (Kirkman, Rosen, Gibson, Tesluk, & McPherson, 2002). When team members understand the goals and behaviors expected of them, subsequent meetings should run more efficiently, as team members should have a better shared understanding, or shared mental model, of their task and team (Cannon-Bowers, Salas, & Converse, 1990). Established goals and expectations also indirectly communicate to team members the behaviors that may be viewed as counterproductive. Thus, meeting facilitators should strongly consider holding an initial face-to-

face team meeting to introduce members of the virtual team, establish goal and role clarity, and develop intra-team rules (Hertel, Geister, & Konradt, 2005).

However, there may be instances where face-to-face initial meetings may simply not be possible, due to travel constraints or budget concerns. Recent literature has begun to point to the value of technology which may mimic social interactions, such as via video conferencing or similar cue-rich media. As highlighted by Kirkman and Mathieu (2005) in their dimensions of team virtuality, one of the critical factors of an effective media is its ability to relay informational value. To this end, some organizations have even requested that all email messages be accompanied by a picture of the person sending the message. This way, the recipient of the meeting invite will know what the sender looks like.

When planning the initial team meeting, facilitators should consider following the guidelines which will be further discussed in detail later (see Table 1), especially if this meeting is being held via videoconference. Facilitators should also consider holding the initial meeting on-site or at a corporate location, because it has been suggested that meeting at project sites or work locations can provide team members with important contextual information (Hinds & Weisband, 2003). For initial videoconference meetings, the facilitators could select a room where some or all of the future meetings will take place for familiarity purposes. Organizations concerned with the financial and or logistical difficulty associated with hosting an initial face-to-face meeting should consider implementing a company intranet (if one has not been created already) that allows employees to view the pictures of their coworkers to decrease ambiguity; several organizations currently provide this feature to their employees, and many organizations use employee photographs on the company intranet in conjunction with several other types of

technologies as a means by which meeting goers may be able to develop a shared sense of connection with others (Rusman, Van Bruggen, Sloep, Valcke, & Koper, 2013).

Guidelines for Successful Facilitation of Virtual Team Meetings

The processes involved in conducting virtual meetings may depend on organizational variables, such as an organization's culture and climate and space availability, as well as on individual variables, such as an individual's cultural background and access to technology. Relatively little research exists regarding virtual team meeting facilitation (Beise, Neiderman, & Bernake, 1999; Mittleman, Briggs, & Nunamaker, 2000; Romano, Nunamaker, & Briggs, 1999). However, there are several other contributing literatures that may offer a unique perspective in order to begin understanding factors that may play a role in these types of settings. Drawing upon meeting and virtual team literatures, as well as the broader teamwork literature, several themes emerged as representing key factors and strategies that may aid in the facilitation of effective global virtual team meetings. Derived from this research base, we next propose a set of six guidelines that offer scientifically driven advice regarding the procedure needed for adequate facilitation (see Table 1). Each guideline is followed by two detailed action steps, and each step along with its corresponding action steps are provided in Table 1 and will be discussed in further detail in the sections to come. It is hoped that this table can provide any reader with a guided checklist for any type of virtual meeting. It is also suspected that the proposed guidelines could serve as a checklist for a face-to-face meeting utilizing electronic information and communication technology.

The six guidelines involve: 1) selecting the virtual team meeting facilitator(s); 2) selecting the appropriate information and communication technology for the meeting; 3) setting meeting norms; 4) setting and reinforcing team roles in meetings; 5) acknowledging time zone

and cultural differences when planning for and conducting virtual team meetings; and 6) following up with action items after the meeting. Although some of these guidelines (e.g., reinforce roles) will actually take place during the meeting itself, facilitators should keep all of these factors in mind when planning the meeting. As a rule of thumb, selecting facilitators, technologies, and considering team member geographic and cultural differences will occur before the meeting, setting and reinforcing team roles will occur during the meeting, and sending out action items will typically occur after the meeting, although this can occur during a meeting as well. Depending on facilitator preferences and time limitations, meeting norms may be

Table 1. Proposed guidelines for successful facilitation of virtual team meetings.

Select a Facilitator	Select Appropriate ICT(s)	Set Meeting Norms	Set and Reinforce Roles	Acknowledge Differences	Follow Up the Meeting with Action Items
Determine facilitator based on team structure / team need	Determine most appropriate ICT(s) for team needs	Discuss meeting times, frequency, and agendas	Develop team charter outlining expected behaviors	Consider team member time zones	Appoint a Scribe who is not the facilitator
Alert individual(s) of his / her (their) role	Select and implement most appropriate ICT(s)	Discuss expected communication and behaviors during meetings	Begin and end meetings discussing role clarity	Consider team member cultural diversity	Utilize appropriate media to facilitate action item communication

discussed with or without the group ahead of time, and, thus, setting meeting norms may occur before or during the meeting.

Guideline 1: Selecting a Facilitator

Selecting a facilitator is imperative to the virtual team meeting process, as the facilitator leads the team toward objectives aimed at accomplishing organizational goals, and it is the duty

of the facilitator to ensure that the meeting goes as smoothly as possible (Kayser, 1994). Previous research on teams suggests that leadership drives team performance, and that this leadership can be represented formally (i.e., the team is led by a supervisor) or informally (i.e., the leadership on the team is shared) within virtual team environments (Shuffler, Wiese, Salas, & Burke, 2010). Indeed, a facilitator may be the leader within the team, or may be an outside consultant, as such consultants may aid in ensuring that members feel that they can be more open in their discussions and comments (Avery, 1981; Niederman & Volkema, 1999). Although facilitators may not necessarily be the leaders of the team, they are expected to take on leadership responsibilities as they manage the situation to the best of their abilities.

Facilitators can be self-selected or chosen by the group or a supervisor. A facilitator may be chosen because of his or her technological expertise, geographical location, level within the company, position on the team, and countless other reasons depending on team need (Niederman & Volkema, 1999). Facilitators may also be chosen due to successful track records of hosting virtual team meetings. More than one facilitator may be chosen to keep the meeting on task, although having too many facilitators for a meeting may prove counterproductive if too many people are trying to manage it. For example, a virtual team comprised of ten employees may only warrant one facilitator, whereas a much larger virtual team meeting comprised of thirty employees may warrant either two or three facilitators, or a skilled facilitator and sophisticated communication technology, such as WebEx[1]. For large virtual meetings where two facilitators are necessary, one facilitator may manage the meeting structure and thinking, whereas the second may be necessary to manage the technical aspects of coordinating the meeting, scheduling participants, and keeping everyone on board from a technology standpoint (Dickson,

[1] WebEx is an information and communication technology designed to improve virtual meetings. https://signup.webex.com/webexmeetings/US/sem_acquisition.html?CPM=KNC-sem&TrackID=1031582&country=US&psearchID=www.webex.com

Partridge, & Robinson, 1993). Only having one type of facilitator or the other may in fact impede in meeting success, as meetings where facilitators only tend to technology can be less successful from an interpersonal standpoint (George, Dennis, & Nunamaker, 1992).

If a meeting will have one facilitator (e.g., the supervisor), the facilitator should strongly consider appointing an individual to take notes and keep track of time during the meeting while the facilitator manages the thinking of the group, keeps the participants on task, and establishes direction during the meeting (DeBono, 1999). Although research exists to suggest that leaders should effectively manage time during meetings (Allen & Rogelberg, 2013), DeBono (1999) also suggests that facilitators who appoint others to manage the meeting time and note may be able to be more actively involved in the meeting process.

If an individual is chosen by others rather than self-selected, it is the responsibility of the team and or the supervisor to alert this individual of his or her responsibility as facilitator for the upcoming meeting. Often, telling the individual the day of the meeting, and in some cases the day before, may hinder the individual's ability to pull together the appropriate technologies, plan the agenda, and send out meeting invitations—in addition to his or her workload. Supervisor facilitators may regularly plan for these reoccurring meetings and allocate more time to planning, suggesting that supervisors should consider facilitating meetings, if applicable, to avoid hassle among team members. Overall, facilitators can play a critical role in the formation and direction of virtual team meetings, and their selection therefore requires extreme care.

Guideline 2: Selecting Information and Communication Technology

Whoever the facilitator may be, one of the first decisions he or she may have to make relates to the use of technology. Facilitators must be adept at selecting and tailoring appropriate technological solutions that conform to their team's communication preferences and meet the purpose of each meeting (Daim et al., 2012). Information and communication technology (ICT)

describes the technology used to relay information to users and facilitate communication between users. Multiple ICTs methods exist, including instant messaging/chat, groupware/shared services, remote access and control, web conferencing, file transfer, email, and telephone (Ebrahim, Ahmed, & Taha, 2009). ICTs are critical to virtual team function and organizations should be prepared to purchase ICTs if choosing to implement one or more virtual teams.

Determining Appropriate ICTs

The first action step toward selecting ICTs is determining which ICTs would be appropriate for one or more upcoming virtual team meetings. As previously discussed, different forms of technology afford different advantages and disadvantages in terms of meeting facilitation effectiveness. Kirkman and Mathieu (2005) highlight three dimensions of virtuality which may influence the selection of appropriate ICTs in meeting situations. First, the degree to which a meeting must rely on virtual tools should be considered—is it appropriate for interactions to occur primarily via virtual means, or are there actions that may need to occur in face to face situations? This may be driven to a great degree by a second dimension, the informational value of the tools being used. Different tools may be better able to provide informational value based on the types of tasks to be accomplished during a meeting. For example, if a team is trying to come up with a design for a new product during a meeting, a collaborative online workspace may be highly beneficial so that members can jointly draw and integrate ideas using a shared visual representation. In this case, videoconferencing may also be useful to supplement and illustrate non-verbal cues of team members, given the creative context. However, for a meeting that simply requires reviewing and approving an agenda for a future meeting, an email with the file and a telephone conference call may suffice. This relates to a third dimension presented by Kirkman and Mathieu (2005), the synchronicity. For meetings,

tools and virtual environments that afford high levels of synchronous interactions may be critical, but being able to asynchronously follow up (e.g., via email or text) or review documentation after a meeting could also be beneficial.

Depending on these different factors, facilitators should carefully select the appropriate media and methods for communicating. For example, if individuals will only need to be able to converse during the meeting, and face-to-face communication is not required, then facilitators will need to access a telephone with teleconferencing features. If a team member must be able to share a computer screen with his or her coworkers during a distributed team meeting, then the facilitator will need to locate a screen sharing tool. As can be expected, the size of the virtual team will have a direct impact on the choice, complexity, and cost of the ICT (Riopelle et al., 2003), and offering a variety of ICTs to a virtual team may help them feel more connected. For example, if three employees from different geographic locations are working on the same spreadsheet in Microsoft Excel at the same time, the employees may feel more connected, as well as more competent, if they all have access to a telephone, a headset, screen sharing, and instant messaging. When it comes time to share the spreadsheet they have created together with each other, they will likely utilize file sharing and email.

Facilitators must also determine when each communication method should be used (e.g., email for status reports, chats for project issues; Cascio, 2000). Timing can be particularly important in virtual environments, especially given the demanding contexts in which many such teams interact (Hertel, Geister, & Konradt, 2005). Selecting the appropriate tools for a given situation may be best done utilizing the dimensions highlighted previously (Kirkman & Mathieu, 2005), with facilitators ensuring that the media selected matches the type and quality of information to be shared (Mesmer-Magnus, DeChurch, Jimenez, Wildman, & Shuffler, 2011).

Based on meta-analytic evidence from Mesmer-Magnus and colleagues (2011) regarding information sharing in virtual environments, meeting tasks that require high amounts of social interaction and openness of information for success should leverage richer media forms (e.g., videoconferencing), while those that require unique information sharing may be fine with less rich media forms (e.g., email). Facilitators should also be prepared to answer questions about ICTs and know who to contact if an ICT stops working. Many organizations now offer information technology support and some even have their own Information Technology business unit; these systems are a step in the right direction toward assisting the millions of virtual teams that operate in workplaces across the globe. However, a lack of understanding or issues with technology can be avoided with proper training for all team members and the implementation of a back-up plan for technology failure (Bal & Foster, 2000; Ebrahim, Ahmed, & Taha, 2009). Training could be conducted via videoconference as well as on-site depending on the needs of the task and the team.

Ultimately, some meetings demand a face-to-face medium. Discussion and resolution of serious team conflicts or crises, as well as sharing of individual negative feedback, should be conducted face-to-face, even if done via videoconference (Holton, 2001). Although employee time is valuable, meetings to discuss team conflicts or crises should be taken seriously, and facilitators or supervisors should set a mandatory meeting time and face-to-face medium for communication. Videoconferencing to discuss individual negative feedback should be held privately between the facilitator and team member in order to avoid unnecessary humiliation, and it is important for facilitators to avoid making socio-technological errors (e.g., accidentally leaving out a team member from the meeting invitation) as often as possible in order to preserve team cohesion and trust.

Implementing the ICTs

The final step to selecting ICTs involves implementing them. Implementation will likely require permission from the company or team supervisor. Facilitators should refrain from being fearful of asking permission to implement ICTs if the ICT is necessary to hold the virtual meeting and or make it more productive. If a facilitator believes that a supervisor may be skeptical about using a particular ICT, he or she could demonstrate the difficulty of conducting the virtual meeting without it in addition to conducting research on the product and various brands. An important point to mention concerns that of ease of use; if one brand of a particular ICT appears more difficult to use, consider finding another. If the software, tool, or service is necessary for group facilitation, then the team could consider offering training to any individuals who may facilitate the team's virtual meetings in the future. Lastly, facilitators should remember that invitations, whether formal or informal, are an important aspect of meetings and facilitators must ensure that every member of the team receives the team meeting invite and agenda.

Guideline 3: Setting Meeting Norms

As may now seem evident, the nature of virtual team meetings requires a more formal structure than for collocated teams, placing a large burden on the team facilitator (Bal & Foster, 2000). Establishing meeting norms early on can help to ensure that the facilitator has adequate time to remain task-focused during meetings, as opposed to tending to housekeeping details. Norms are defined as "standards that regulate group members' behavior" (Forsyth, 1999, p. 121). In virtual settings, norms can be critical for establishing guidelines of effective task procedures and modes of communication and coordination (Maynard, Mathieu, Rapp, & Gilson, 2012). Establishing norms early on will help team members understand not only what they are expected to work on, but how they are expected to accomplish that work as a team. A virtual team

meeting facilitator should set specific guidelines for meeting times, frequency, agendas, and, when appropriate, guidelines for how and when team members should communicate (Kayworth & Leidner, 2000). Tending to housekeeping details early on may enable facilitators to avoid dealing with them in the future.

Meeting Timing

The establishment of a meeting time should not be hastily overlooked. Facilitators, or those scheduling the meeting on behalf of a facilitator, should exercise courtesy when forming a meeting time. Many employees participate in meetings several hours each day. Scheduling a 30 minute meeting versus a 60 minute meeting may make a significant difference in one's ability to accomplish his or her tasks that particular workday. Electronic meeting schedule tools can be utilized to save several email exchanges and to ensure one is not scheduling a meeting on top of another participant's scheduled meeting. Although attempting to accommodate busy schedules may sometimes (albeit rarely) lead to meeting times outside of typical work hours, this should be avoided as much as possible as it interferes with work-life balance (Greenhaus, Collinis, & Shaw, 2003). Parents may find it difficult to attend before- and after-hours meeting because of child responsibilities. Further, employees who commute long hours to work may prefer meetings to occur at least thirty minutes after the work day begins to have the opportunity to get organized.

A facilitator should also only schedule a meeting for as long as he or she believes it will take to accomplish the purpose of the meeting. If this individual strongly believes the meeting would need to take 90 minutes to identify and plan for a solution to a problem, then that facilitator should schedule the meeting for 90 minutes. Otherwise, the team may find itself

needing to schedule a follow-up meeting—and perhaps another follow-up meeting—which may prove counterproductive for all participants involved.

In general, certain times will work better for others, and geographic location certainly plays a major role in determining a best time to host a virtual meeting. As we will discuss later, the timing of meetings may be further challenged when individuals from different geographic time zones are required to participate in the meeting (Connaughton & Shuffler, 2007). Thus, setting expectations and encouraging courteous behaviors when scheduling meetings may be an extremely important norm for virtual meetings. For example, although meeting time management (i.e., the extent to which managers start meetings on time, end when scheduled to end, and schedule meetings with adequate time for employees to arrange their other activities for the day; Allen & Rogelberg, 2013) is considered to be an important element of successful virtual team facilitation by some researchers, it is expected that some employees will be late to meetings due to work-related or personal reasons. Facilitator should exercise kindness to team members entering the meeting late and quickly catch the team member up to speed. Because many professionals schedule meetings back-to-back, meetings tend to be affected by previous meetings.. Although this is not the ideal situation for meetings in which important outcomes are relied on, but experience in the organizational setting reveals that lateness is a common occurrence within meetings and one that would be difficult to eradicate. Similarly, it is common courtesy that employees can leave the meeting at its scheduled time of conclusion—even if that employee arrived late. Research by Kahn (1990) indicates that effective time management behaviors, whether before, during, or at conclusion of a meeting, communicate to employees that their work is meaningful and that they are cared about and respected.

Meeting Frequency

Another factor to consider when establishing meeting norms is meeting frequency, or how often meetings will occur. Considering how often employees may be in meetings, and that a large number of workgroup meetings has been found to be negatively related to overall employee well-being (Rogelburg, Leach, Warr, & Burnfield, 2006), facilitators should create meetings only when necessary. Once a virtual project team is put in place, determining how often the virtual team should meet will depend on how often meetings are required to progress team performance. Facilitators should exercise caution in holding multiple meetings within a week, as too many interactions may affect membership within other teams, and may decrease work productivity all together (O'Leary, Mortensen, & Woolley, 2011). Certain individuals will not be able to make frequent team meetings due to their work load, and facilitators will need to work around these challenges in order to ensure that these individuals can participate in the team meeting when needed. In order to accomplish these tasks, facilitators may find it helpful to survey members of the team about their preferences for meeting frequency (as well as meeting time)

Meeting Agenda Norms

For each meeting, facilitators must ensure that agendas are set and communicated ahead of time, agendas are followed and team members stay on task, and action items are developed to maintain momentum and progress until the next meeting (Malhotra, Majchrzak, & Rosen, 2007). During each meeting, facilitators must also control sidebar conversations, ensure that no one dominates the conversation, provide opportunities for introverts to contribute, remain completely focused, and follow up post-meeting as needed to ensure understanding (Hambley, O'Neill, & Kline, 2007).

How the facilitator communicates the agenda and keeps team members on task may depend on the status and roles of the team members (Earley & Mosakowski, 2000). If the virtual team has a formal team leader, then the facilitator of a given team meeting may be a supervisor or a supervisor's direct report. How a direct report handles a meeting may differ greatly from how a supervisor leads a meeting. (Employees facing ambiguity around these issues could ask their supervisor for guidance on how to appropriately facilitate a meeting for a given situation in that organization.) On the other hand, virtual teams characterized by shared leadership functions may lack a formal, participative leader (Shuffler, Wiese, Burke, & Salas, 2010), and the facilitator may change from meeting to meeting depending on the topic at hand. Because virtual team settings are distributed to varying degrees (e.g., partially and fully), professionals should consider the context and content of the meeting, as well as the individuals' expertise and geographically location, when developing agendas.

Communication and Behavior Norms

In regards to communicating, it is important that facilitators aim to establish appropriate ground rules during the meeting, while also ensuring that they incorporate the communication and behavior norms driven by individual participant needs (Von Glinow, et al., 2004). As will further be discussed in the following section, individuals may bring their own preferences for communicating and engaging with others, and it is important that norms are created that both accept these values and preferences, while also creating a "hybrid culture" of communicating that works for all team members together (Earley & Mosakowski, 2000). The establishment of common courtesy behaviors and communication norms and rules may even help to reduce the overall costs associated with ineffective meetings that detract from employee productivity and organizational success (Rogelberg, Shanock, & Scott, 2012). Indeed, factors such as

dysfunctional communication such as criticizing and complaining others in meetings have been found to reduce team productivity, while problem focused, action oriented communication can be highly beneficial to meeting success (Kauffeld & Lehmann-Willenbrock, 2012). Thus, appropriate communication ground rules may include factors such as focusing communication on the task and problems to be solved as opposed to socio-emotional communication more oriented towards negativity or complaining; they may also include the need for individuals to speak up and share their uniquely held information as opposed to focusing on facts that are already known across such groups, as this type of unique information is critical to, yet not shared for, successful group outcomes (Mesmer-Magnus, et al., 2010).

While important, creating effective communication and behavior norms may be somewhat challenging in virtual meetings, given the cultural differences that may often occur in such environments. Indeed, such cultural differences may impact how individuals from different backgrounds may expect such meetings to run and the behaviors that are considered acceptable (Von Glinow, et al., 2004). For example, Kohler and colleagues (2012) found that for meetings, Germans and U.S. Americans hold very different expectations regarding the purpose, content, structure, and timing of meetings, as well as the roles of participants in such meetings. As communication and behavioral norms may vary by culture, facilitators should take the time during the initial meeting to both understand and highlight the behaviors that will be viewed as appropriate and advantageous during meetings. It is critical however that facilitators do not necessarily favor a single culture, but instead work to incorporate norms, values, and beliefs regarding communication that complement the range of cultures involved. Earley and Mosakowski (2000) make a case for this in their work regarding such hybrid culture development in teams, arguing that the successful implementation of these integrated cultures

can lead to positive interactions and feelings of inclusion that support the overall efficacy and success of the group in the longer term. Furthermore, having a shared understanding of what may be viewed as valuable and inappropriate behavioral and communication norms may aid in reducing conflict and misunderstandings across collaborators in a virtual meeting environment (Meyer, 1993).

Guideline 4: Setting and Reinforcing Team Roles

Sometimes, one of the biggest pitfalls that can occur in any team is lack of role clarity in team settings. Role clarification is critical for the effective coordination of actions in teams, defined as the enactment of behavioral mechanisms necessary to perform a task and transform team resources into outcomes (Sims & Salas, 2007). Coordination can take several forms, as individuals within a team may be performing the same or complementary tasks which may range in regards to their interdependence (Guastello & Guastello, 1998). Furthermore, coordination can be explicit, where team members intentionally utilize mechanisms such as planning and communication to manage interdependencies, or implicit, whereby team members anticipate team needs and dynamically adjust their behaviors accordingly without having to be instructed (Rico, Sanchez-Manzanares, Gil, & Gibson, 2008).

Given the importance of role clarification in team settings, this idea may also prove to be useful in virtual meeting environments. To prevent conflict and stress in the future, facilitators may aid in improving the outcomes of such meetings by clarifying team goals and ensuring that team members know their individual roles when they leave the meeting, in order to ensure effective future coordination. Indeed, operating in virtual environments may be particularly used by their ability to bring together individuals from a range of areas of expertise and functional backgrounds (Connaughton & Shuffler, 2007). Thus, in order to maximize the unique contributions and knowledge of individual members, making roles clear may help to build

relationships and networks among members that are necessary for overall meeting and team performance (Crawford & LePine, 2012). Therefore, it is recommended that facilitators set and reinforce team roles by developing a team charter and commencing and concluding each meeting with a discussion of role clarity.

Developing a Team Charter

Team charters can be highly advantageous to ensuring role clarity during virtual team meetings, as they outline expected team member behaviors and provide a means for documenting team norms (Mathieu & Rapp, 2009). Because team charters document a team's goals and responsibilities, and often specify ground rules and deliverables, charters help to reinforce established meeting norms by holding team members accountable for their actions. Indeed, Mathieu and Rapp (2009) found that high quality team charters that are established early in the life of a team can aid in facilitating effective team performance strategies later on. Although team members can hold each other accountable for breaches of the team charter, individual roles of team members must be mutually decided and agreed upon (Fisher, Rayner, & Belgard, 1995). Thus, we recommend that teams develop their team charter during the first meeting and, if not, by the conclusion of the second meeting in order to guide team behaviors in a similar fashion. Facilitators should make sure they or another coworker comes prepared to begin outlining the charter during the meeting and record meeting minutes and notes to be included in the charter at a later time. This charter should be brought to all future meetings and adhered to by leadership overtime; otherwise, team members may assume that they will not be held accountable for their actions, and the team contract may become null and void. Charters should also be continually reviewed, with members being held up to the standards and consequences held in these charters to ensure their effectiveness (Asencio, Carter, & DeChurch, Zaccaro, & Fiore, 2012).

Clarifying Team Roles

The same colleague who disclosed with us the importance of clarity in teams also mentioned the understated and underestimated impact that role clarity can have on team performance; this colleague later commented that lack of role clarity created confusion surrounding work projects and tension and mistrust between coworkers at his workplace. According to these suggestions, team members aware of their specific roles and contributions to the team may need less attention from management and may be more productive, and there is evidence to suggest that reinforcing role assignments has positive outcomes on team process (Holton, 2001). Specifically, Holton (2001) suggests that role assignments be reinforced by beginning each meeting with team members declaring their assigned roles, and ending each meeting with a review and agreement that all roles have been adequately covered.

The supervisor described in the brief case description presented at the beginning of this chapter facilitated a virtual team meeting in textbook fashion. In line with our recommendations, this supervisor began the meeting by having team members address the group with perceptions of their individual roles and found time toward the end of the meeting to clarify next steps. Time appeared to be on this team's side that morning.

However, time is not always on everyone's side. In fact, staying within the allotted meeting time for some teams proves impossible. If a meeting is projected to run over time, the facilitator may not have an opportunity to allow members to reinforce and declare assigned roles. Going over meeting time is a common occurrence in organizational settings, and, if reestablishing team goals and member roles is imperative for a team to function, this team's facilitator should consider adjusting meeting times to fit this necessity. Team members may

show signs of disapproval during the moment, but may be happy they stayed and clarified roles and deliverables after the fact.

Facilitators may also consider holding individual virtual meetings with team members to check in on individual assignments (Kirkman, Rosen, Gibson, Tesluk, & McPherson, 2002). And, because modern work often requires employees to be flexible in adapting to dynamic, technological environments, it is likely that team member roles may change over time. Thus, facilitators should be aware of changes in the workgroup that may affect how coworkers interact with one another. For example, if two coworkers have switched roles, or a new employee has joined the team, the facilitator should strive to keep updated on this kind of information, for significant changes to task structure may require significant preplanning.

In sum, working to clarify roles can be an important aspect of virtual team meetings. Providing tools such as the use of team charters or the follow up of larger team meetings with individual team meetings to ensure a shared understanding of the goals and tasks can aid in facilitating improved communication and information sharing in virtual teams, which in turn may lead to improved outcomes (Mesmer-Magnus, et al., 2010). However, it is important to keep in mind that this may not be necessary for every type of team, as in some situations roles may need to be more fluid, such as if there is a great amount of turnover or multiple membership of members that requires the ability to change or switch responsibilities based on workload (O'Leary, Mortensen, & Woolley, 2011). Further, facilitators may want to keep in mind that a forced role structure may induce hierarchies that could impede members working together or sharing leadership responsibilities, a critical component for team success (D'Innocenzo, Mathieu, & Kukenberger, 2014). Thus, facilitators should carefully evaluate the degree to which

role clarification is necessary for the meetings and/or more distal team performance, and plan accordingly given the possible tools suggested.

Guideline 5: Acknowledging Team Member Differences

Last but not least, two contributors of successful virtual team meeting facilitation concern that of acknowledging time zones and cultural differences. Time can serve as both an advantage and disadvantage to virtual teams. Asynchronous work over multiple time zones provides greater opportunity for productivity, but, as previously mentioned, asynchronous work may also extend the workday and create difficulty with coordination of work and meetings (Saunders, Van Slyke, & Vogel, 2004). Time visions (i.e., different perceptions of time across sets of time dimensions) vary based on country, longitude, culture, and religion; norms must be set for meeting etiquette, scheduling, attendance, punctuality, and feedback based on the time visions of all team members. To accommodate team members in various time zones, it is suggested to rotate meeting times so that everyone 'shares the pain' of inconvenience (Hambley, O'Neill, & Kline, 2007). Currently, many global virtual teams coordinate time-zone differences by alternating having a meeting conducive to a particular time zone. Etiquette still applies to virtual meetings, however; if the majority of members—including the team leader—are collocated in one time-zone, facilitators should consider holding the meeting at an hour that is acceptable for them. Facilitators should also conduct team meetings with clients in the client's prospective time zone, unless the client specifically says otherwise.

Cultural differences may also impact the facilitation of virtual meetings in numerous ways. In seeking to provide guidance to organizations there has been a fair amount of work conducted which examines intracultural differences in group or team-based work. For example, Lundby and Jolton (2010) note that research has demonstrated the impact cultural differences

have on several group processes, including cooperation (Kirkman & Shapiro, 2001), communication and feedback (Earley et al., 1999), conflict type (Elron, 1997), cohesion (Man & Lam, 2003), team efficacy (Gibson, 1999), adaptation (Harrison, McKinnon, Wu, & Chow, 2000), decision making (Kirchmeyer et al., 1992), and team performance (Gibson, 1999; Matveev & Nelson, 2004). However, Lundby and Jolton (2010) comment that many organizations are increasingly relying on multi-cultural teams, but may be overlooking the challenges associated with leading and functioning in a team comprised of individuals with diverse backgrounds, traditions, motivations, and concerns (Dinwoodie, 2005).

When considering language barriers, multiple works suggest that a combination of email, teleconferencing, and videoconferencing is an effective ICT strategy for culturally diverse teams, as non-video media allow for non-native speakers to more comfortably express their ideas. Similarly, Grosse (2002) found that non-native English speakers were more willing to type comments in a meeting using virtual software as compared to speaking during face-to-face meetings or teleconferencing. Facilitators must be adept to signs of discomfort from team members and be willing to forgo norms that apply to only one country during global team meetings. If an individual appears uncomfortable during a face-to-face meeting, or is not as participative during a teleconference as he or she typically is in-person, the facilitator might consider following up with this individual after the meeting to ensure that the virtual team meeting's form of communication is not hindering this individual from participating and voicing opinions.

Asynchronous methods also remove the time pressure of synchronous communication, so that individuals can take longer to ensure message clarity, to understand the context of a message and respond appropriately, and to overcome language barriers (Warkentin, Sayeed, &

Hightower, 1997). Despite cultural differences that may exist, research has demonstrated that multicultural team members often try to be 'on their best behavior' when engaging with team members because they view themselves as a representative of their country and culture (Elron & Vigoda, 2003). Facilitators can also ensure that diversity of background and expertise is understood and appreciated by sharing information about team members to inform them about individual training, experience and professional affiliations (Malhotra, Majchrzak, & Rosen, 2007; Tyran, Tyran, & Shepherd, 2003).

Follow Up the Meeting with Action Items

When a virtual meeting adjourns, participants will move on to other tasks or meetings. Without a memory aid, some team members may forget important content that was discussed, especially given the high demand of meetings that many employees experience on a regular basis (Luong, et al., 2005). Following up each meeting with action items that give team members a chance refresh their memories as well as provide them with tangible next steps orients the team toward the same goals and may save team members from having to unnecessarily email the facilitator (or other team members) afterward.

Appoint a Scribe

During meetings, a lot of information may be communicated. In virtual meetings, it may be particularly critical to capture the unique information shared among team members, especially since this information may be coming via a range of media varying in quality (Kirkman & Mathieu, 2005). For example, a virtual team holding a web conference meeting may relay information via video chatting that offers both verbal and non-verbal cues, screen sharing of documents and programs, slideshows, emails, and chats, all of which may provide a vast amount of information that needs to be documented for future purposes (Mesmer-Magnus, et al., 2011).

Thus, it is particularly critical that an appropriate set of minutes are developed for each meeting via a scribe (Leach, et al., 2009). As previously mentioned, facilitators may select an individual to manage time for the team's meeting. The facilitator may also ask this individual to take notes during the meeting in order to streamline the meeting process. This note taker has the ability to capture and preserve important information for his or her team, and facilitators should seriously consider appointing an individual to be a scribe during each meeting. The same individual does not need to be the scribe each time, unless the team feels that this is necessary or appropriate. The scribe may participate in the meeting, but his or her main purpose is to record the thoughts, ideas, and solutions (if any) of team members to be turned into action items once the meeting has adjourned. The criticality of such scribes and the resulting minutes has been exhibited in traditional face-to-face settings, whereby such meeting design features have been found to facilitate meeting effectiveness (Leach, et al., 2009). Given that more information may be communicated in virtual meetings due to the introduction of a wider range of media, this guideline is expected to be just as beneficial and perhaps even more so in these settings.

Utilize Appropriate Media to Facilitate Action Item Communication

As discussed throughout this chapter, one of the most notable differences regarding virtual team meetings is the use of technology to facilitate interactions. Thus, it is equally as important to ensure that the appropriate media is used to communicate action items following the conclusion of a virtual meeting. After the meeting, the facilitator and scribe may choose the next steps for following up on action items discussed during the meeting. Action items (i.e., units, tasks, or actions) that can be handled by a single person) included on a meeting agenda should be discussed during a meeting and assigned to team members post-meeting (Leach, et al., 2009; Spencer & Pruss, 1992). In order to compose action items, a facilitator may ask the scribe to

organize the information and email it to the facilitator, who will reorganize and enhance the meeting notes to create items based on his or her own reflections of the meeting. In other settings or situations, a facilitator may ask the scribe to email out action items based on the scribe's meeting notes. Action items (i.e., units, tasks, or actions) that can be handled by a single person) can be discussed during a meeting and assigned to team members post-meeting. Although action items serve an important purpose for moving forward any team project, actions items may be especially important for virtual teams because virtual team meetings may not produce decisions at the same rate that collocated team meetings do. In a multi-corporate study of 600 employees, 80% reported that a major challenge they experienced in a virtual team meeting was speed of decision making, and 76% reported that another major challenge with virtual team meetings centered on the method of decision making (Virtual Teams Survey Report, 2010). Based on these findings, virtual teams may benefit from action items post-meeting because it enables team members to accomplish actions on their own without having to spend time on these actions during valuable meeting time.

Certainly, while action items are critical for virtual team meetings, the other important factor is the means by which such action items are communicated and followed up with to ensure completion. As noted by Kirkman and Mathieu (2005), virtual teams may operate with a range of media that possess varying levels of appropriateness for communicating different forms of information. While action items could be followed up with a simple email message to all team members, longer emails may not be fully read or followed through by team members due to the lower information quality offered by this medium (Warkentin, et al., 1997). A lack of attention to meeting follow up may be further exaggerated by the virtual nature of the team, especially if the team does not meet very often. However, this may also depend upon factors such as the

composition of the team, as more conscientious members may be more likely to follow up regardless of the media communicate (Bell. 2007). Regardless of the team composition, to ensure meeting follow up with action items, facilitators may need to select a more dynamic media that matches the team structure and needs. This may be in the form of a support system that enables members to recognize their specific tasks, respond to others either synchronously or asyncrhonosly, and to mark when tasks have been completed (Warkentin & Beranek, 1999; Warkentin, et al., 1997). Overall, the appropriate selection of such tools that match team needs should help to facilitate the communication and follow through on action items.

Future Research Directions

Throughout this chapter, we have offered several guidelines and recommendations regarding the facilitation of virtual meetings based on the current state of the science. While there is much to be gleaned from existing literature, there are also several important areas that require future exploration in order to continue to advance the science. First, in terms of preparing for initial meetings, additional research is needed to explore the appropriate media and formats that may be as conducive to holding such meetings in comparison to face-to-face settings. As previously discussed, while holding initial meetings in a collocated setting can be highly advantageous, from a practical standpoint this may not be possible in all situations. Therefore, it is important that researchers better understand how media may be best utilized to promote successful social interactions necessary for positive meeting outcomes.

Additionally, another area for future exploration involves the selection of facilitators, particularly the need to utilize multiple facilitators to meet the differing needs that may occur in a virtual environment. Given the challenging demands of interacting in virtual environments, both in terms of task and social components, there may be a need for multiple facilitators, or even the

rotation of meeting facilitators based upon the needs of the meeting participants given particular agenda issues (Zaccaro & DeChurch, 2011). While still a new area of study, there have been promising findings supporting the idea that collective leadership can facilitate effective teamwork and performance (Carson, Tesluk, & Marrone, 2007). Collective leadership and facilitation of meetings may, therefore, be even more appropriate than a single facilitator or leader, as having multiple team members take on leadership needs can aid in ensuring team needs are being met throughout the duration of a meeting (Day, Gronn, & Salas, 2006). However, further research is necessary to better understand this area.

A third and final area of research relates to the exploration of the impact of organizational culture and norms on the structuring and facilitation of virtual meetings. While typical cultural research regarding teams and groups tends to focus on national culture differences, there also may be cultural dynamics in regards to organizational differences (Connaughton & Shuffler, 2007). As organizations are becoming increasingly collaborative and utilize technology to manage joint projects, differences in perceptions regarding how meetings should be conducted may be driven based not only on nationality differences, but also differences in organizational values, beliefs and norms (Keyton, 2011). Organizational culture is a multilevel system of artifacts, values and assumptions, and may also involve subcultures based on different organizational subcomponents, such as different units or departments (Schein, 2004). From a meeting perspective, organizations may place differential values on factors such as being open and accepting of technology, seeing the value of face to face meetings, or being willing to share facilitation responsibilities. This may in turn affect how individuals participate in meetings as representatives of their respective organizations or organizational units, and may be particularly challenging to establishing effective norms and roles among meeting attendees. Indeed, such

differences may create faultlines that may further exacerbate issues and conflicts that will prevent successful future interactions (Li & Hambrick, 2005). Thus, understanding not only national culture but also organizational cultural influences may be particularly important and relevant for our future exploration of successful meeting facilitation in virtual contexts.

Concluding Remarks

Certainly, virtual team meetings are often far from perfect. Connecting over time and space using various forms of media to connect can be very challenging, especially if attendees are from very different cultures and backgrounds with their own unique approaches to meeting design and management (Kohler, et al., 2012). However, as in our initial opening example, with appropriate tools and interventions, it is very possible for virtual teams to conduct successful meetings, initially and over their lifespan, in order to lead to eventual team success. Virtual teams comprise much of the modern organizational landscape and research demonstrates that the facilitation of virtual team meetings can have major consequences on team outcomes. Facilitators who take the time to plan for these specific types of meetings by selecting appropriate people and technologies to do the job and get the job done, ensuring that team roles are clearly communicated and followed, and integrating team member diversity will likely experience positive meeting outcomes.

We do add the caveat that while our guidelines are derived from existing scientific research and best practices regarding teamwork, meetings, and virtual work, there may be exceptions to these guidelines depending on the specific situational constraints for a given team and facilitator. For example, in some settings there may not be an option to select a facilitator based on the team needs; in such cases, it may be important then for the selected facilitator to attend to the team needs and design a meeting approach that works given the situation and that

facilitators strengths and abilities. However, we do believe that these recommendations and guidelines offered to provide a scientifically driven, solid starting point that can help to reduce the stress and burden of developing, implementing, and following up with virtual team meetings. Further, it is imperative that additional research is conducted in order to better understand the unique nuances and requirements of successful virtual team meetings, so that these guidelines can be better refined and developed over the course of time and further changes in technology.

Although some study has been devoted to virtual team meetings, there is still much to learn about how team meetings should be run and how they impact performance over time. We hope that the recommendations and guidelines offered in this chapter can be adapted to fit the needs of various virtual team situations, and that they will be refined as further research uncovers the nuances of virtual meeting facilitation in a range of settings. Further, staying current on ICTs and embracing the opportunities technology can offer may help facilitators accomplish their objectives and contribute to the promotion of modern and sophisticated work environments. In sum, while there is still much to be explored regarding the set up and facilitation of effective virtual team meetings, we hope that the insight offered here will aid in providing some initial directions to both research and practice.

References

Alge, B. J., Wiethoff, C., & Klein, H. J. (2003). When does the medium matter? Knowledge-building experiences and opportunities in decision-making teams. *Organizational Behavior and Human Decision Processes, 91*, 26-37.

Allen, J. A., & Rogelburg, S. G. (2013). Manager-led group meetings: A context for promoting employee engagement. *Group and Organizational Management, 38*(5), 543-569. doi: 10.1177/1059601113503040

Asencio, R., Carter, D. R., DeChurch, L. A., Zaccaro, S. J., & Fiore, S. M. (2012). Charting a course for collaboration: A multiteam perspective. *Translational Behavioral Medicine*, *2*(4), 487-494.

Avery, M. (1981). *Building united judgment: A handbook for consensus decision making.* Madison, Wisconsin: Center for Conflict Resolution.

Bal, J., & Foster, P. (2000). Managing the virtual team and controlling effectiveness. *International Journal of Production Research, 38*(17), 4019-4032. doi.org/10.1080/00207540050204885

Beise, C., Evaristo, R., & Niederman, F. (2003, January). Virtual meetings and tasks: from GSS to DGSS to project management. In *System Sciences, 2003. Proceedings of the 36th Annual Hawaii International Conference on* (pp. 9). IEEE.

Burke, C. S., Shuffler, M. L., Salas, E., & Gelfand, M. (2010). Multicultural teams: Critical team processes and guidelines. In K. Lundby & J. Jolton (Eds.), *Going global: Practical applications and recommendations for HR and OD Professionals in the global workplace* (pp. 46-83). San Francisco, CA: Jossey-Bass.

Cannon-Bowers, J. A., Salas, E., & Converse, S. A. (1990). Cognitive psychology and team training: Shared mental models in complex systems. *Human Factors Bulletin, 33*, 1–4.

Cascio, W. F. (2000). Managing a virtual workplace. *The Academy of Management Executive, 14*(3), 81-90. doi.org/10.5465/AME.2000.4468068

Cascio, W. F., & Shurygailo, S. (2003). E-leadership and virtual teams. *Organizational Dynamics, 31*, 362-376.

Cogliser, C. C., Gardner, W., Trank, C. Q., Gavin, M., Halbesleben, J., & Seers, A. (2013). Not all group exchange structures are created equal effects of forms and levels of exchange on work outcomes in virtual teams. *Journal of Leadership & Organizational Studies*, *20*(2), 242-251.

Connaughton, S.L. & Shuffler, M.L. (2007). Multinational/multicultural distributed teams: A review and future research agenda. *Small Group Research, 38*(3), 387-412.

Coutu, D. L. (1998). Trust in virtual teams. *Harvard Business Review, 76*(3), 20–21.

Crawford, E., & LePine, J. (2012). A configural theory of team processes: Accounting for the structure of taskwork and teamwork. *Academy of Management Review*, amr-2011.

Culnan, M. J., & Markus, M. L. (1987). Information technologies. In F. M. Jablin, L. L. Putnam, K. H. Roberts, & L. W. Porter (Eds.), *Handbook of organizational communication: An interdisciplinary perspective* (pp. 420–443). Newbury Park, CA: Sage.

D'Innocenzo, L., Mathieu, J. E., & Kukenberger, M. R. (2014). A meta-analysis of different forms of shared leadership–team performance relations. *Journal of Management*, 0149206314525205.

Daim, T. U., Ha, A., Reutiman, S., Hughes, B., Pathak, U., Bynum, W., & Bhatla, A. (2012). Exploring the communication breakdown in global virtual teams. *International Journal of Project Management, 30*(2), 199-212. doi.org/10.1016/j.ijproman.2011.06.004

Day, D. V., Gronn, P., & Salas, E. (2006). Leadership in team-based organizations: On the threshold of a new era. *Leadership Quarterly, 17*, 211−216.

DeBono, E. (1999). Six Thinking Hats®. First Back Bay: MICA Management Resources.

Dinwoodie, D. L. (2005). Solving the dilemma: A leader's guide to managing diversity. *Leadership in Action*, *25*(2), 3-6.

Earley, C. P., & Mosakowski, E. (2000). Creating hybrid team cultures: An empirical test of transnational team functioning. *Academy of Management Journal*, *43*(1), 26-49.

Earley, P. C., Gibson, C. B., & Chen, C. C. (1999). "How did I do?" versus "how did we do?" Cultural contrasts of performance feedback use and self-efficacy. *Journal of Cross-Cultural Psychology*, *30*(5), 594-619.

Ebrahim, N. A., Ahmed, S., & Taha, Z. (2009). Virtual teams: A literature review. *Australian Journal of Basic and Applied Sciences, 3*(3), 2653-2669.

Elron, E. 1997. Top management teams within multinational corporations: Effects of cultural heterogeneity. *Leadership Quarterly, 8,* 393–412.

Elron, E., & Vigoda, E. (2003). Influence and political processes in virtual teams. In C. B. Gibson & S. G. Cohen (Eds.), *Virtual teams that work: Creating conditions for virtual team effectiveness* (pp. 317-334). San Francisco, CA: Jossey-Bass.

Fisher, K., Rayner, S., Belgard, W., & Armstrong, J. (1995). Tips for teams: A ready reference for solving common team problems. New York: McGraw-Hill.

Forsyth, D. R. (1999). *Group dynamics* (3rd ed.). Belmont, CA: Brooks/Cole.

Gibson & S. G. Cohen (Eds.), *Virtual teams that work: Creating conditions for virtual team effectiveness* (pp. 403–421). San Francisco: Jossey-Bass.

Gibson, C. B. (1999). Do they do what they believe they can? Group efficacy and group effectiveness across tasks and cultures. *Academy of Management Journal*, *42*(2), 138-152.

Gibson, C. B., & Cohen, S. G. (2003). The last word: Conclusions and implications. In C. B.

Grosse, C. U. (2002). Managing communication within virtual intercultural teams. *Business Communication Quarterly, 65*(4), 22-38. doi.org/10.1177/108056990206500404

Guastello, S. J., & Guastello, D. D. (1998). Origins of coordination and team effectiveness: A perspective from game theory and nonlinear dynamics. *Journal of Applied Psychology, 83*, 423-437.

Hambley, L. A., O'Neill, T. A., & Kline, T. J. B. (2007). Virtual team leadership: The effects of leadership style and communication medium on team interaction styles and outcomes. *Organizational Behavior and Human Decision Processes, 103*(1)*,* 1-20. doi.org/10.1016/j.obhdp.2006.09.004

Harrison, G. L., McKinnon, J. L., Wu, A., & Chow, C. W. (2000). Cultural influences on adaptation to fluid workgroups and teams. *Journal of International Business Studies*, *31*(3), 489-505.

Henttonen, K., & Blomqvist, K. (2005). Managing distance in a global virtual team: The evolution of trust through technology-mediated relational communication. *Strategic Change*, *14*(2), 107-119. doi.org/10.1002/jsc.714

Hertel, G., Geister, S., & Konradt, U. (2005). Managing virtual teams: A review of current empirical research. *Human Resource Management Review, 15,* 69-95. doi:10.1016/j.hrmr.2005.01.002

Hinds, P. J., & Weisband, S. P. (2003). Knowledge sharing and shared understanding in virtual teams. In C. B. Gibson & S. G. Cohen (Eds.), *Virtual teams that work: Creating conditions for virtual team effectiveness* (pp. 21-36). San Francisco, CA: Jossey-Bass.

Holton, J. A. (2001). Building trust and collaboration in a virtual team. *Team Performance Management: An International Journal, 7*(3/4), 36-47. doi.org/10.1108/13527590110395621

Kahn, W. A. (1990). Psychological conditions of personal engagement and disengagement at work. *Academy of Management Journal, 33*(4), 692-724. doi:10.2307/256287

Karsh, B. T. (2004). Beyond usability: designing effective technology implementation systems to promote patient safety. Quality and Safety in Health Care, 13, 388-394. doi 10.1136/qshc.2004.010322

Kauffeld, S. & Lehmann-Willenbrock, N. (2012). Meetings matter: Effects of team meetings on team and organizational Success. *Small Group Research*, *43*(2), 130–158.

Kauffeld, S., & Lehmann-Willenbrock, N. (2012). Meetings Matter Effects of Team Meetings on Team and Organizational Success. *Small Group Research*,*43*(2), 130-158.

Kayworth, T., & Leidner, D. (2000). The global virtual manager: A prescription for success. *European Management Journal, 18*(2), 183-194. doi.org/10.1016/S0263-2373(99)0009000

Keyton, J. (2011). *Communication and organizational culture: A key to understanding work experiences* (2nd ed.). Thousand Oaks, CA: Sage.

Kirchmeyer, C., & Cohen, A. (1992). Multicultural Groups Their Performance and Reactions with Constructive Conflict. *Group & Organization Management*,*17*(2), 153-170.

Kirkman, B. L., & Mathieu, J. E. (2005). The dimensions and antecedents of team virtuality. *Journal of Management, 31*(5), 700-718. doi:10.1177/0149206305279113

Kirkman, B. L., & Shapiro, D. L. (2001). The impact of cultural values on job satisfaction and organizational commitment in self-managing work teams: The mediating role of employee resistance. *Academy of Management Journal*, *44*(3), 557-569.

Kirkman, B. L., Rosen, B., Gibson, C. B., Tesluk, P. E., & McPherson, S. O. (2002). Five challenges to virtual team success: Lessons from Sabre, Inc. *The Academy of Management Executive, 16*(3), 67-79. doi:10.5465/AME.2002.8540322

Köhler, T., Cramton, C. D., & Hinds, P. J. (2012). The Meeting Genre Across Cultures: Insights From Three German–American Collaborations. *Small Group Research*, 1046496411429600.

Leach, D. J., Rogelberg, S. G., Warr, P. B., & Burnfield, J. L. (2009). Perceived meeting effectiveness: the role of design characteristics. *Journal of Business and Psychology, 24(*1), 65-76.

Li, J., & Hambrick, D. C. (2005). Factional groups: A new vantage on demographic faultlines, conflict, and disintegration in work teams. *Academy of Management Journal*, *48*(5), 794-813.

Lundby, K. & Jolton, J. (2010). *Going global: Practical applications and recommendations for HR and OD professionals in the global workplace*. San Francisco, CA: John Wiley & Sons.

Lundby, K., & Jolton, J. (2010). *Going global: Practical applications and recommendations for hr and od professionals in the global workplace*. San Francisco, CA: Jossey-Bass.

Malhotra, A. Majchrzak, A., & Rosen, B. (2007). Leading virtual teams. *Academy of Management Perspectives, 21*(1), 60-70.

Man, D. C., & Lam, S. S. (2003). The effects of job complexity and autonomy on cohesiveness in collectivistic and individualistic work groups: a cross-cultural analysis. *Journal of Organizational Behavior*, *24*(8), 979-1001.

Mathieu, J. E., & Rapp, T. L. (2009). Laying the foundation for successful team performance trajectories: The roles of team charters and performance strategies. *Journal of Applied Psychology*, *94*(1), 90.

Matveev, A. V., & Nelson, P. E. (2004). Cross cultural communication competence and multicultural team performance perceptions of American and Russian managers. *International Journal of Cross Cultural Management*, *4*(2), 253-270.

Maynard, M. T., Mathieu, J. E., Rapp, T. L., & Gilson, L. L. (2012). Something(s) old and something(s) new: Modeling drivers of global virtual team effectiveness. *Journal of Organizational Behavior, 33*(3), 342-365.

Mesmer-Magnus, J. R., DeChurch, L. A., Jimenez, M. J., Wildman, J., & Shuffler, M. (2011). A meta-analytic examination of virtuality and information sharing in teams. *Organizational Behavior and Human Decision Processes, 115*, 214-225.

Meyer, H.-D. (1993). The cultural gap in long-term international work groups: A German-American case study. *European Management Journal, 11*, 93-101. doi:10.1016/0263-2373(93)90029-h

Mittleman, D. D., Briggs, R. O., Nunamaker, J. F., Jr., & Romano, N.C. (2000). Best practices in facilitating virtual meetings: Some notes from initial experiences. *Group Facilitation: A Research & Applications Journal, 2, 6-15.*

Niederman, F., & Volkema, R. (1999). The effects of facilitator characteristics on meeting preparation, set up and implementation. *Small Group Research, 30(3),* 330-360.

Ocker, R., Fjermestad, J., Hiltz, S. R., & Johnson, K. (1998). Effects of four modes of group communication on the outcomes of software requirements determination. *Journal of Management Information Systems, 15*(1), 99-118.

O'Leary, M. B., Mortensen, M., & Woolley, A. W. (2011). Multiple team membership: A theoretical model of its effects on productivity and learning for individuals and teams. *Academy of Management Review*, *36*(3), 461-478.

O'Leary, M. B., Mortensen, M., & Woolley, A. W. (2011). Multiple team membership: A theoretical model of its effects on productivity and learning for individuals and teams. *Academy of Management Review*, *36*(3), 461-478.

Olson, G. M. & Olson, J. S. (2002). *Groupware and computer-supported cooperative work. The human-computer interaction handbook: Fundamentals, evolving technologies and emerging applications.* (pp. 583-595). Hillsdale, NJ: L. Erlbaum Associates Inc.

Pauleen, D., & Yoong, P. (2001). Facilitating virtual team relationships via internet and conventional communication channels. *Internet Research: Electronic Networking Applications and Policies, 11*(3), 190-202. doi.org/10.1108/10662240110396450

Powell, A., Piccoli, G., & Ives, B. (2004). Virtual teams: A review of current literature and directions for future research. *The DATA BASE for Advances in Information Systems, 35*(1), 6-36. doi.org/10.1145/968464.968467

Rice, R. E., & Love, G. (1987). Electronic emotion: Socioemotional content in a computer-mediated network. *Communication Research, 14,* 85–108. doi.org/10.1177/009365087014001005

Rico, R., Sanchez-Manzanares, M., Gil, F., and Gibson, C. B. (2008). Team coordination processes: A team knowledge-based approach. Academy of Management Review, 33(1), 163-185.

Riopelle, K., Gluesing, J. C., Alcordo, T. C., Baba, M., Britt, D., McKether, W., Monplaisir, L., Ratner, H. H., & Wagner, K. H. (2003). Context, task, and the evolution of technology use in global virtual teams. In C. B. Gibson & S. G. Cohen (Eds.), *Virtual teams that work: Creating conditions for virtual team effectiveness* (pp. 239-264). San Francisco, CA: Jossey-Bass.

Rogelberg, S. G., Leach, D. J., Warr, P. B., & Burnfield, J. L. (2006). 'Not another meeting!' Are meeting time demands related to employee well-being? *Journal of Applied Psychology, 91*, 86-96.

Rogelberg, S. G., Shanock, L. R., & Scott, C. W. (2011). Wasted time and money in meetings: Increasing return on investment. *Small Group Research*, 1046496411429170.

Romano, N. C., Nunamaker, J. F. Jr., & Briggs, R. O. (1999). Distributed GSS facilitation and participation: Field action research. Proceedings of the 32nd Annual Hawaii International Conference on Systems Sciences. Hawaii, IEEE.

Rusman, E., van Bruggen, J., Sloep, P. B., Valcke, M., & Koper, R. (2013). The mind's eye on personal profiles: A cognitive perspective on profile elements that inform initial trustworthiness assessments and social awareness in virtual project teams. *Computer Supported Cooperative Work, 22,*159-179.

Sanchez, R., & Mahoney, J. T. (1996). Modularity, flexibility, and knowledge management in product and organization design. *Strategic Management Journal, 71*, 63-76.

Saunders, C., Van Slyke, C., & Vogel, D. R. (2004). My time or yours? Managing time visions in global virtual teams. *The Academy of Management Executive, 18*(1), 19-31. doi.org/10.5465/AME.2004.12691177

Shuffler, M.L., Wiese, C. W., Salas, E., & Burke, C. S., (2010). Leading one another across time and space: Exploring shared leadership functions in virtual teams. *Journal of Work and Organizational Psychology, 26*(1), 3-17.

Sims D. E., & Salas, E. (2007). When teams fail in organizations: what creates teamwork breakdowns? In J. Langan-Fox, C. L. Cooper, & R. J. Klimoski (Eds), *Research companion to the dysfunctional workplace: Management challenges and symptoms* (pp. 302-317). Cheltenham, United Kingdom: Edward Elgar. Publishing Ltd; 2007:302–317.

Spencer, J., & Pruss, A. (1992). *Managing your team: How to organise people for maximum results*. London: Piatkus.

Suchan, J., & Hayzak, G. (2001). The communication characteristics of virtual teams: A case study. *IEEE Transactions on Professional Communication, 44*(3), 174-186.

Tyran, K. L., Tyran, C. K., & Shepherd, M. (2003). Exploring emerging leadership in virtual teams. In C. B. Gibson & S. G. Cohen (Eds.), *Virtual teams that work: Creating conditions for virtual team effectiveness* (pp. 183-195). San Francisco, CA: Jossey-Bass.

Von Glinow, M. A., Shapiro, D. I., & Brett, J. M. (2004). Can we talk and should we? Managing emotional conflict in multicultural teams. *Academy of Management Review, 29*(4), 578-592.

Warkentin, M. E., & Beranek, P. M. (1999). Training to improve virtual team communication. *Information Systems Journal, 9*, 271-289.

Warkentin, M. E., Sayeed, L., & Hightower, R. (1997). Virtual teams versus face-to-face teams: An exploratory study of a web-based conference system. *Decision Sciences*, *28*(4), 975-996. doi:10.1111/j.15405915.1997.tb01338.x

Wiesenfeld, B. M., Raghuram, S., & Garud, R. (1999). Communication patterns as determinants of organizational identification in a virtual organization. *Organization Science, 10*, 777–790.

Zaccaro, S. J., & DeChurch, L. A. (2011). Leadership forms and functions in multiteam systems. In S. J. Zaccaro, M. A. Marks, & L. A. DeChurch (eds.), *Multiteam systems: An organizational form for dynamic and complex environments*. Taylor & Francis.

Author Biographies

Brooke B. Allison, M.S., is a doctoral student at Clemson University pursuing a Ph.D. in Industrial-Organizational Psychology, where she also completed her Master's degree. Brooke has previously authored a scholarly book chapter and has a variety of research and applied experience, including the inquiry into multiteam systems, team communication, work-family balance, and well-being, as well as two summers of performance management and leader succession work at TIAA-CREF. She has earned the Bernard Caffrey Award and the Academic Excellence Award for Psychology during her time at Clemson.

Marissa L. Shuffler, PhD, has over ten years of experience conducting basic and applied research in the areas of team development, leadership, and organizational effectiveness. Dr. Shuffler is an Assistant Professor of Industrial/Organizational Psychology at Clemson University. Her areas of expertise include team and leader training and development, intercultural collaboration, multi-team systems, and decision-making/adaptation, with an emphasis on high risk and complex environments. Dr. Shuffler has served as scientific task lead on several major interdisciplinary research projects, including the empirical investigation of key leadership issues in NASA long duration space missions, and the design of training interventions aimed at improving multiteam system coordination and functioning. Dr. Shuffler conducted this and similar research for government, military, and industry, including the U.S. Army Research Institute, the Department of Labor, the Center for Army Leadership, the Department of Homeland Security, the U.S. Air Force, TIAA-CREF, and the Four Seasons Hotel. Her work to date includes over 40 publications and over 75 presentations. She holds a master's in I/O psychology from George Mason University, and doctorate in I/O psychology from the University of Central Florida.

Allison Wallace, M.S., is a doctoral candidate in Clemson University's Industrial-Organizational Psychology program, where she also completed her Master's degree. Her research areas of interest include individual and group judgment and decision making, leadership development, executive coaching and assessment, and mergers and acquisitions. Allison has spent the past three years contributing to both organizational and leadership development initiatives at multiple Fortune 100 companies. She has held several consultant roles in the areas of educational assessment, workplace engagement, and high-inference data coding, and enjoys teaching at the undergraduate level at two universities. Allison looks forward to her future career as an I/O consultant.